Cyno proteome half-life pipeline — Reproducibility Runbook (v2)

Kosugi, Youhei — Preclinical Biomarkers, Takeda

Reproducible pack for Copilot Cowork

Table of Contents

Cyno proteome half-life pipeline — Reproducibility Runbook (v2)

0. Why v1 (the PDF alone) is not enough

Cyno proteome half-life pipeline — Reproducibility Runbook (v2)

Owner: Kosugi, Youhei (Director, Preclinical Biomarkers, Takeda)
Supersedes: Cyno_pipeline_runbook.pdf (v1, prose-only methods guide)
Purpose: survive a Copilot Cowork refresh (which wipes input/ and working/) and reproduce Tier 1-1 … 1-6 **numerically**, not just methodologically.

---

0. Why v1 (the PDF alone) is not enough

The v1 PDF is an excellent *methods* description, but a prose guide cannot guarantee that the **same numbers** come back, because:

1. **The analysis code was never saved.** v1 keeps scripts in working/ and only final outputs in output/; a refresh wipes working/. Re-deriving code from prose drifts on details the prose does not pin (exact feature set, the ~50 complex-name regexes, rank tie-handling, the permutation sampling universe).
2. **The published feature set is *smaller* than the prose list.** v1 Tier 1-4 lists MW, GRAVY, instability index, TM count, etc., but the *actual* model that produced $R^2 = 0.0190$ used only: log10(length), pI, disorder fraction, PEST max, and three N-end dummies. Building from the prose list does **not** reproduce $R^2 = 0.0190$.
3. **Live databases drift.** v1 fetches UniProt features / GO / DisGeNET at run time; those change daily, so identical code gives different numbers on a later date unless the pulled annotations are frozen.
4. **Underspecified knobs + stale intermediates.** Examples this pack uncovered: the permutation null in Tier 1-5 samples from the **whole cohort** (not the member subset) — unstated in v1; the Tier 1-1 outlier set needs a coverage cut the prose omits; and the saved Tier 1-1 pair

table was built on the **older 3,515-row cohort**, so it still carries the ambiguous Mrc1;Slc39A12 row that the rev result excludes. All three are now pinned/handled in code.

This pack closes 1–4 by shipping the code, the pinned inputs, and frozen snapshots of every external dependency, plus an acceptance table.

---

1. What a Cowork refresh keeps vs. wipes

Location	Refresh behaviour	Contents
output/ → OneDrive Documents/Cowork	**synced out; recoverable**	results, figures, this pack, the derived cohort
input/	**wiped**	raw uploads: Monkey_half-life.xlsx, corum_humanComplexes.txt, Cyno.fasta, DIAMOND pairs.tsv
working/	**wiped**	analysis scripts + intermediates (this is where v1's code was lost)

> ⚠ The pack survives **only because it lives in output/**. Keep a local copy too — if a refresh is ever configured to also clear the OneDrive Cowork folder, the synced copy is your only backup. Confirm your refresh policy.

---

2. Pack layout

```
cyno_repro_pack/
├── REPRODUCIBILITY_RUNBOOK.md   ← this file
├── RECOVERY.md                  ← 4-step post-refresh quick start
├── ENVIRONMENT.md               ← Python/NumPy versions, seed, no-network note
├── MANIFEST.md                  ← SHA-256 checksums + acceptance (expected) table
├── code/                        ← NumPy-only re-implementation (the durable artifact)
│   ├── stats_numpy.py           ← ranks, Spearman, Mann-Whitney, Cliff's δ, OLS, t-p
│   ├── family_lib.py            ← shared seeded engine for the family-dispersion analysis
│   ├── tier1_1_paralog_outliers.py
│   ├── tier1_1b_family_dispersion_sweep.py   ← identity sweep 20–80
│   ├── tier1_2_repeat_class.py
│   ├── tier1_4_seq_regression.py
```

```
│   ├── tier1_5_corum_membership.py
│   ├── tier1_6_stoich_regul.py
│   ├── fig_repeat_class.py      ← Fig_repeat_class_halflife_rev.png (days)
│   ├── fig_seq_features.py      ← Fig_seq_features_rev.png (days)
│   ├── fig_family_dispersion.py ← Fig_family_dispersion_threshold_sweep_rev.png
(seeded)
│   ├── fig_identity_boxplot.py  ← Fig_identity_vs_halflife_boxplot_rev.png (+
Kruskal–Wallis)
│   └── run_all.py               ← runs every tier + prints expected vs recomputed
├── data/
│   ├── inputs/                  ← pinned raw inputs (wiped on refresh otherwise)
│   │   ├── cyno_target_rev.tsv        (cohort, n = 3,514)
│   │   ├── cyno_target_rev.fasta
│   │   ├── corum_humanComplexes.txt   (CORUM 5.x, Human)
│   │   └── cyno_pairs.tsv             (DIAMOND all-vs-all)
│   ├── snapshots/               ← FROZEN external-data captures (no live fetch needed)
│   │   ├── seq_features_rev.tsv           (UniProt-derived features per protein)
│   │   ├── Repeat_class_halflife_rev.tsv   (UniProt domain → repeat class)
│   │   ├── corum_member_table_rev.tsv      (member / non-member, promiscuity)
│   │   ├── corum_strat_assignment_rev.tsv  (NONE/STOICH/REGUL/BOTH)
│   │   ├── corum_complex_stats_rev.tsv     (the 481-complex set, per-complex SD)
│   │   └── pair_table_rev.tsv             (11,527 evaluated paralog pairs)
│   └── expected/                ← acceptance targets (the published rev outputs)
└── results/                     ← regenerated by run_all.py (compare to expected/)
```

The **snapshots** are the key idea: they capture the live UniProt / domain / CORUM state from the original run, so the pipeline is reproduced **offline and date-stable**.

---

3. How to reproduce after a refresh

See RECOVERY.md for the short version. In full:

1. Re-open Cowork; the pack is in your OneDrive Documents/Cowork. Upload the whole cyno_repro_pack/ folder back into the session (or point Copilot at it).
2. cd code && python3 run_all.py (no network needed; NumPy only).
3. Compare results/ to data/expected/ and to the acceptance table in MANIFEST.md. Deterministic tiers must match to the stated precision; the one permutation p must land within Monte-Carlo error.
4. To run a **new species** (Tier 1-7 prep), see §6.

---

4. Per-tier reproduction status

Validated in-sandbox against the pinned data/expected/ files. “Exact” means the recomputed value equals the published value to the reported precision.

Tier	Headline result	Reproduction status
1-1 Paralog outliers	top pair Srsf6/Srsf4, fold 9.22	**Exact (31/31)** — from the frozen evaluated-pairs table pair_table_rev.tsv + two recovered rules (see note below). A *naive* re-run from raw DIAMOND diverges (24–35); the saved table is the key.
1-1b Family dispersion sweep	sequence-similar paralogs are co-stable (perm P < 1e-3)	**Families + within-family SD exact**; identity sweep extended to **20, 30, 40, 50, 60, 70, 80**. From pair_table_rev.tsv (connected components, cov ≥ 50%, size ≥ 3). Recovered details: graph from the curated pair table (not raw DIAMOND), within-family SD uses **population SD (ddof = 0)**. Proteins-in-families differ by ≤ 1 at low identity (stored graph carried one extra near-boundary edge); random_SD/perm_p are Monte-Carlo (seed = 42).
1-2 Repeat-class	ALL_REPEAT n=148, **p = 1.984e-2**	**Exact** — every family n, median, percentiles, and p reproduced.
1-3 GO comparison	(BH-FDR over GO terms)	**Not reproducible offline** — outputs were never saved and it needs live UniProt-GOA. Re-run requires a GOA snapshot (see §5).
1-4 Sequence	whole-model **R^2 =**	**Exact** — R^2, every β,

Tier	Headline result	Reproduction status
regression	**0.0190**; pI β=+0.0380, p=1.07e-9	SE, p, and all four Spearman ρ reproduced.
1-5 CORUM membership	within-complex perm **p = 0.63**; member vs non-member null	**Exact (deterministic) + within MC error (permutation)** — members 1442 / non 2072, medians 7.69/8.77, MW p=7.3e-6, 481 complexes, observed SD 1.0702, random 1.0632 (pub 1.0630), perm p 0.631 (pub 0.6271).
1-6 STOICH vs REGUL	**STOICH vs REGUL p = 0.92**	**Exact** — class sizes 127/1132/183/2072, all medians, and every Mann-Whitney p reproduced.

Bottom line: every statistical tier (1-1, 1-2, 1-4, 1-5, 1-6) reproduces from this pack — the null findings that motivate Tier 1-7 (sequence, complex, and structural-complex predictors all fail) hold exactly. Only Tier 1-3 (GO) is out, for lack of a saved GOA snapshot.

Three reproducibility lessons this pack made concrete

- **Tier 1-4 feature set** — must come from the frozen seq_features_rev.tsv, not the prose feature list, to land on R^2 = 0.0190.
- **Tier 1-5 permutation pool** — the size-matched null draws from the **whole cohort (3,514)**, not the 1,442 members. Member pool → null 1.073, p≈0.44; whole-cohort pool → 1.063, p≈0.63 (matching 0.6271). One unstated choice.
- **Tier 1-1 needs the saved intermediate + a cohort-version fix.** Rebuilding the pair table from raw DIAMOND diverges (24–35 pairs) because the coverage/HSP/ best-hit rules aren't in the prose. The original pair_table_rev.tsv (11,527 evaluated pairs with pident/cov_min/Δlog2) was however saved — and it was built on the **pre-dedup 3,515-row cohort**, so it still carries the ambiguous Mrc1;Slc39A12 composite. Re-applying the rev-cohort gene filter drops exactly that stale self-pair, and pident≥80 + 2-fold + cov_min≥0.5 then yields exactly the 31 published pairs. *Nothing in the data changed* — the apparent non-reproducibility was a stale intermediate + an unrecorded coverage cutoff.

5. External dependencies and how they are pinned

Dependency	Original source	How it is frozen here
Half-life cohort (n=3,514)	derived from Monkey_half-life.xlsx	data/inputs/cyno_target_rev.tsv (the **derived** cohort, after dropping the ambiguous Mrc1;Slc39A12 row, 3,515→3,514)
UniProt sequence features	live UniProt at run time	data/snapshots/seq_features_rev.tsv
Repeat-domain class	live UniProt domains	data/snapshots/Repeat_class_halflife_rev.tsv
CORUM complexes	corum_humanComplexes.txt (CORUM 5.x)	pinned raw file **and** the collapsed 481-complex set corum_complex_stats_rev.tsv
DIAMOND paralog pairs	DIAMOND all-vs-all (binary not in sandbox)	raw data/inputs/cyno_pairs.tsv **and** the evaluated-pairs intermediate data/snapshots/pair_table_rev.tsv (this is what makes Tier 1-1 reproducible)
GO terms (Tier 1-3)	live UniProt-GOA	**not captured** — Tier 1-3 cannot be reproduced offline until a GOA snapshot is saved
DisGeNET/OMIM (Tier 1-1 disease tags)	live	not captured; not required for the null findings

Gap to close for full coverage: snapshot UniProt-GOA (GO terms per accession) the next time Tier 1-3 is run, and record the CORUM release + UniProt release dates.

6. Running a new species (Tier 1-7 prep)

The pack reproduces the *cyno* result. For mouse/rat, supply the same-format inputs and the pipeline re-runs:

1. input/<species>_target.tsv — columns gene, alias, accession, length, half_life_h (see v1 §1.1). The data-prep step that builds this from a raw half-life table is **not yet scripted** — add build_cohort.py documenting the xlsx→tsv mapping and the exact dedup rule so the *upstream* step is reproducible too.
2. input/corum_humanComplexes.txt — reuse the pinned copy for cross-species ortholog mapping (uppercase gene symbol key).
3. input/<species>_pairs.tsv — DIAMOND all-vs-all, produced **outside** the sandbox (DIAMOND is not installable here).
4. For new species, snapshots cannot be reused — UniProt features / domains / GO must be fetched live (needs network) and then **frozen** the same way, so the species result becomes reproducible in turn.

---

7. Reproducibility rules (carried from v1, now enforced in code)

- All permutation statistics use numpy.default_rng(42).
- Statistics depend on **NumPy only**; the custom rank / Mann-Whitney / Spearman / OLS / t-p implementations in stats_numpy.py are authoritative (do not swap in SciPy).
- Half-life is handled internally in **hours**; figures/legends convert to days (day = h/24). log2 for effect sizes, log10 for correlations.
- Redundant / peptide-ambiguous rows (e.g. Mrc1;Slc39A12) are excluded by accession; only confident gene assignments are used (rev cohort: 3,515 → 3,514).
- Cross-species comparisons key on **uppercase gene symbol**, with UniProt accession as the secondary key.

RECOVERY — restore the Cyno pipeline after a Cowork refresh

A refresh wipes input/ and working/. This pack lives in output/ and syncs to your OneDrive Documents/Cowork, so it survives. Four steps to reproduce:

1. **Re-upload the pack.** In the refreshed Cowork session, bring the whole cyno_repro_pack/ folder back from OneDrive Documents/Cowork into the session.

2. **Run everything (no network needed):**

```
cd cyno_repro_pack/code
python3 run_all.py
```

For a fast check (≈1 min) use `N_PERM=2000 python3 run_all.py`; the canonical run uses `N_PERM=10000` for the Tier 1-5 permutation (a few minutes).

3. **Verify.** Each tier prints `recomputed (expected …)`. Confirm against the acceptance table in `MANIFEST.md`. The headline checks:
    - Tier 1-4 whole-model **R^2 = 0.0190**
    - Tier 1-5 within-complex permutation **p ≈ 0.63**
    - Tier 1-6 STOICH vs REGUL **p = 0.92**
    - Tier 1-2 ALL_REPEAT **p = 1.984e-2**
4. **(Optional) check file integrity:**

```
cd cyno_repro_pack
sha256sum -c MANIFEST.sha256
```

That's it — no UniProt, CORUM, or GOA lookups are required; every external input is frozen under `data/snapshots/`.

- Tier 1-1 paralog outliers **31/31** (0 extra, 0 missing)

Only Tier 1-3 (GO comparison) is not included — its outputs were never saved and it needs a live UniProt-GOA snapshot. See `REPRODUCIBILITY_RUNBOOK.md` §5.

Environment manifest — Cyno proteome half-life pipeline

This pack was validated in the Copilot Cowork sandbox with the following runtime. Reproduction is robust to minor version drift, but record this for provenance.

Component	Version used	Notes
Python	3.12.9	CPython
NumPy	2.5.0	the **only** required third-party package
matplotlib	3.11.0	figures only; not needed for the statistics
pandas	3.0.3	present but **not used** by the pipeline
SciPy	**absent**	intentionally — all stats

Component	Version used	Notes
		are NumPy-only re-implementations

Hard requirements

- **NumPy only** for every statistical result. code/stats_numpy.py re-implements ranks, Spearman, Mann-Whitney (tie-corrected), Cliff's delta, OLS, and the Student-t p-value (via a regularized incomplete-beta evaluation) so that **no SciPy is required**. Do not "upgrade" to SciPy — results are pinned to these implementations.
- **Determinism**: every permutation uses numpy.default_rng(42). Seed = 42.
- **No network access is needed.** All external data (UniProt features, domain calls, GO terms, CORUM membership) is captured in data/snapshots/. The pipeline never re-fetches a live database, so results do not drift with the state of UniProt / CORUM / GOA on the day you run it.

Tools that are NOT available in the sandbox (and are not needed here)

- **DIAMOND** (the BLAST-like aligner used to build cyno_pairs.tsv). It is a compiled binary and cannot be installed. The DIAMOND output is therefore pinned as data/inputs/cyno_pairs.tsv; for a brand-new species you must produce the pairs file outside the sandbox and upload it.
- pip install / apt — package installation is blocked. Work with NumPy only.

Permutation runtime

Tier 1-5's within-complex SD permutation is the only slow step. With the canonical N_PERM = 10000 and the whole-cohort pool it takes a few minutes in the sandbox. For a quick sanity check use N_PERM=2000 (≈45 s); the permutation p has Monte-Carlo noise of order ±0.01 at that size, which does not change any conclusion.

MANIFEST — acceptance targets & integrity

- **File integrity:** sha256sum -c MANIFEST.sha256 covers the STABLE artifacts (code + pinned data + docs). results/ are regenerated by run_all.py and excluded by design.
- **Determinism:** every permutation uses numpy.default_rng(42).
- **Validated runtime:** Python 3.12.9 / NumPy 2.5.0, no SciPy, no network (see ENVIRONMENT.md).

Acceptance table (recomputed vs published)

A reproduction passes if each value below matches. Deterministic rows must match to the printed precision; the single permutation row must fall within Monte-Carlo error (±~0.01 at N_PERM=10000).

Tier 1-4 — sequence-feature regression (data/expected/Supp_Table_seq_features_rev.tsv)

Quantity	Published	Recomputed	Status
whole-model R^2	0.0190	0.0190	☑ exact
pI β (p)	+0.0380 (1.07e-9)	+0.0380 (1.19e-9)	☑
disorder β (p)	−0.0294 (1.11e-6)	−0.0294 (1.16e-6)	☑
log10(length) β (p)	−0.0051 (0.404)	−0.0051 (0.404)	☑ exact
N-end TYPE1 β (p)	+0.1628 (4.41e-2)	+0.1628 (4.42e-2)	☑
N-end NDEG β (p)	+0.1933 (4.68e-2)	+0.1933 (4.69e-2)	☑
Spearman pI ρ	+0.1083	+0.1083	☑ exact

Tier 1-5 — CORUM membership (data/expected/corum_summary_rev.tsv)

Quantity	Published	Recomputed	Status
members / non-members	1442 / 2072	1442 / 2072	☑ exact
median h member / non	7.6920 / 8.7744	7.6920 / 8.7744	☑ exact
member-vs-non MW p	7.3099e-6	7.3134e-6	☑
complexes (n≥2)	481	481	☑ exact
within-complex SD observed	1.0702	1.0702	☑ exact
within-complex SD random	1.0630	1.0633	☑ (MC)
within-complex perm p	0.6271	0.6290	☑ (MC)
promiscuity ρ (p)	−0.0556 (3.47e-2)	−0.0556 (3.47e-2)	☑ exact

Quantity	Published	Recomputed	Status
complex-size ρ (p)	−0.0774 (8.98e-2)	−0.0774 (8.98e-2)	☑ exact

Tier 1-6 — STOICH vs REGUL (data/expected/corum_strat_summary_rev.tsv)

Quantity	Published	Recomputed	Status
class sizes NONE/STOICH/REGUL/BOTH	2072/127/1132/183	2072/127/1132/183	☑ exact
medians (h)	8.7744/7.6388/8.0829/5.9703	8.7744/7.6388/8.0830/5.9703	☑ exact
STOICH vs REGUL p	9.1616e-1	9.1626e-1	☑
BOTH vs NONE p	2.8169e-9	2.8179e-9	☑

Tier 1-2 — repeat-class (data/expected/Repeat_class_summary_rev.tsv)

Quantity	Published	Recomputed	Status
ALL_REPEAT n / median / p	148 / 6.807 / 1.984e-2	148 / 6.807 / 1.984e-2	☑ exact
TPR n / median / p	36 / 6.786 / 2.617e-2	36 / 6.786 / 2.617e-2	☑ exact
WD40 n / median / p	41 / 6.201 / 7.588e-2	41 / 6.201 / 7.588e-2	☑ exact

Tier 1-1 — paralog outliers (data/expected/Outlier_pairs_id80_2fold_rev.tsv)

From frozen pair_table_rev.tsv + rev-cohort filter + pident≥80, 2-fold, cov_min≥0.5. | Quantity | Published | Recomputed | Status | |—|—|—|—| | outlier pairs | 31 | 31 (0 extra, 0 missing) | ☑ exact | | top pair (Srsf6/Srsf4) | fold 9.22, dlog2 3.205 | fold 9.22, dlog2 3.205 | ☑ exact |

Tier 1-1b — family-dispersion identity sweep (data/expected/Table_family_threshold_sweep_rev.tsv)

From pair_table_rev.tsv; population SD (ddof=0); sweep extended to **20 & 40**. Families and within-family SD reproduce exactly; proteins ≤1 off at low identity; random-set SD / permutation P are Monte-Carlo (seed=42). | Identity | Families | within-SD | Status | |—|—|—|—| | ≥30 | 166 | 0.211 | ☑ | | ≥60 | 59 | 0.138 | ☑ | | ≥70 | 32 | 0.097 | ☑ | | ≥80 | 11 | 0.139 | ☑ | | **≥20 (new)** | 188 | 0.211 | added | | **≥40 (new)** | 125 | 0.185 | added |

Tier 1-3 — GO comparison

Not reproducible offline (outputs never saved; needs live UniProt-GOA). See REPRODUCIBILITY_RUNBOOK.md §5.

File inventory

- code/ (13 files): NumPy-only stats + family_lib.py + tier scripts + run_all.py + 4 figure scripts.
- data/inputs/ (4): pinned cohort, FASTA, CORUM, DIAMOND pairs.
- data/snapshots/ (6): frozen external-data captures (incl. pair_table_rev.tsv).
- data/expected/ (8): published acceptance targets.
- results/ (6): regenerated by run_all.py.